Continuous low-noise confinement without a probability grid

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AI-assisted research note. The argument and exact scalar checks were developed with AI assistance and reviewed separately. The result below is conditional on its stated analytic and entry premises; it is not a proof of the full sharp iid Berry–Esseen conjecture.

1 Conditional statement and acquired application

Let $n \geq 42$ be an actual integer. Let (P, H) be a fixed-target maximizing pair with nonnegative excess $\mathfrak{a}$ over $c = C_{\mathrm{E}}$, where $X \sim P$ is centered and has variance one. Retain the closed-support contact theorem and the true predecessor bound $C_{n-1} \leq c$. The actual raw threshold H can have either sign. Assume an actual central gap $(-A, B)$ with $A, B > 0$, and write

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad a = m/q, \quad b = m/p, \quad \ell = a + b,$$

$$X = \ell(Y - p), \qquad Y = \mathcal{B} + U, \qquad \mathcal{B} = \mathbf{1}_{\{X>0\}}, \qquad \mathbb{E}(U \mid \mathcal{B}) = 0.$$

Define U_i as the conditional value of U on $\{\mathcal{B} = i\}$. Assume the already acquired bounds

$$p \in [2/5, 3/5], \qquad |U| < 3/5, \qquad d = 1 - (A + B)/\ell < 9/125, \qquad \lambda = n\mathbb{E}U^2 \leq 3/40. \tag{1}$$

The gap implies $A \leq a$, $B \leq b$, and hence $d \geq 0$.

Theorem 1. *Under these hypotheses, the entire conditional supports satisfy $|U| < 1/2$. The conclusion is uniform in all raw H and all $n \geq 42$. It includes the closed noise endpoint $\lambda = 3/40$ and zero noise.*

The probability interval in this abstract lemma is $[0.4, 0.6]$. The currently acquired entry supplies the incoming strip and inward bounds only on the union of panels 40–52, namely $[0.40, 0.53]$. The application below is restricted to that smaller interval. In particular this note does not acquire the strip on $(0.53, 0.6]$ or treat the larger-noise region.

2 One common moment enclosure

Put $v = pq$, $s = \mathbb{E}U^2 = \lambda/n$, and $\theta = \mathbb{E}(X - \mathbb{E}[X \mid \mathcal{B}])^2$. Conditional centering and variance give the exact identities

$$\ell^2 = (v + s)^{-1}, \qquad \theta = \frac{s}{v + s} = \frac{\lambda}{nv + \lambda}, \qquad \ell^2 = \frac{1 - \theta}{v}. \tag{2}$$

Throughout (1), $6/25 \leq v \leq 1/4$ and $\max(p, q) \leq 3/5$. Use the rational constants

$$\Theta = \frac{37}{5000}, \qquad E = \frac{249}{125}, \qquad r = \frac{221}{200}.$$

Then

$$\theta \leq \frac{5}{677} < \Theta, \qquad \ell^2 \geq 4(1 - \Theta) > E^2,$$

with exact differences $\Theta - 5/677 = 49/3385000$ and $4(1-\Theta) - E^2 = 73/31250$.

Because $\mathcal{B}$ is the actual sign of X, the absolute cubic expansion has no sign approximation:

$$\rho := \mathbb{E}|X|^3 = \ell^3 \left\{ v(1-2v) + 3\mathbb{E}[|\mathcal{B}-p|U^2] + \mathbb{E}[\operatorname{sgn}(\mathcal{B}-p)U^3] \right\}. \tag{3}$$

The linear term vanishes by conditional centering. The incoming strip, used before the conclusion, gives $|U|^3 \le (3/5)U^2$ and $|\mathcal{B}-p| \le 3/5$. Also $\ell^3 \le v^{-3/2}$. Therefore

$$\rho \le \rho_B(v) + \frac{12/5}{v^{3/2}}\frac{\lambda}{n}, \qquad \rho_B(v) = \frac{1-2v}{\sqrt{v}}.$$

The function ρ_B decreases on $(0, 1/4]$, since its derivative is $-\frac{1}{2}v^{-3/2} - v^{-1/2} < 0$. The exact squared comparison

$$24^2(6/25)^3 - (12/5)^2 = \frac{34416}{15625} > 0$$

gives $\rho \le \rho_B(6/25) + 24\lambda/n < r$. For the last step,

$$r - \frac{3}{70} = \frac{1487}{1400} > 0, \qquad \left(r - \frac{3}{70}\right)^2 \frac{6}{25} - \left(\frac{13}{25}\right)^2 = \frac{8707}{24500000} > 0.$$

If $\theta_\pm$ are the two nonnegative conditional variance contributions in physical coordinates, their sum is θ. Expanding the signed second moment gives

$$\beta := \mathbb{E}[X|X|] = (1-\theta)(1-2p) + \theta_+ - \theta_-.$$

Consequently, for the actual p, including $p > 1/2$,

$$1 - 2p - 2(1-p)\Theta \le \beta \le 1 - 2p + 2p\Theta. \tag{4}$$

All these bounds concern one common marginal. No assertion about independent conditional laws or a moment relaxation is used.

3 All-threshold contact and exterior rays

For $w = n-1$, $s_n = \sqrt{n/w}$, $F_w(t) = \Phi(t/\sqrt{w})$ and $e_n = n(s_n - 1) - 1/2$, fixed-target contact and the predecessor bound give, at every closed-support point t,

$$0 \le \mathcal{H}_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - n\mathfrak{a}, \tag{5}$$

where

$$\frac{\mathcal{H}_n(H,t)}{n^{3/2}} = F_w(H-t) - F_n(H) + tF_n'(H) - \frac{t^2-1}{2}F_n''(H).$$

The term $-n\mathfrak{a}$ is nonpositive and may be dropped. This is a fixed-target statement; no equality between c and the actual score is assumed.

Expand the first term spatially through degree four. Expand the other three terms in variance through orders two, one and one, using $\partial_w F_w = F_w''/2$. At $t = \pm u$, cancellations give

$$\frac{\mathcal{H}_n(H,\pm u)}{s_n^3} = \pm A_u(1-z^2)\phi(z) + \frac{H_4(u)}{24\sqrt{w}}\phi'''(z) + R,$$

$$z = H/\sqrt{w}, \quad A_u = (u^3 - 3u)/6, \qquad H_4(u) = u^4 - 6u^2 + 3.$$

The absolute bounds for the four remainders are

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120w}, \quad \frac{\|\phi^{(5)}\|_\infty}{48w^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8w}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16w^{3/2}}.$$

Gaussian Fourier inversion, integrating absolute powers of frequency, gives $\|\phi^{(4)}\|_\infty \le 3\phi(0) < 6/5$ and $\|\phi^{(5)}\|_\infty \le 8/\pi < 51/20$. For $u \ge 1$ their sum is at most

$$\frac{u^5/100 + 3u/20}{w} + \frac{51(3u^2-2)}{960w^{3/2}}.$$

Thus the remainder extends below the earlier $u \ge 5/2$ presentation. For $u > \sqrt{3}$, $A_u > 0$, and the elementary bounds $\|(1-z^2)\phi(z)\|_\infty = \phi(0)$ and $\|\phi'''\|_\infty \le 2/\pi < 2\phi(0)$ control both signs.

Set $S = 81/80$, $c_0 = 819/2000 < c$ and $e = 8/875$. For $n \ge 42$, $s_n < S$, $\sqrt{w} > 6$, and $e_n < e$. For completeness, $ne_n = s_n^2(2s_n+1)/(2(s_n+1)^2)$ is increasing for $s_n \ge 1$; its value at S is $793881/2073680 < 48/125$. Define

$$E_u = u^5/4100 + 3u/820 + 17(3u^2-2)/78720,$$
$$Q_{r,b}(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + er, \tag{6}$$
$$P_{r,b}(u) = \tfrac{2}{5}S^3\{A_u + |H_4(u)|/72\} + S^3E_u + c_0Q_{r,b}(u). \tag{7}$$

If the directional signed moment is at most b, $Q < 0$ licenses replacing the positive multiplier c by c_0. Then $P < 0$ contradicts (5) at that boundary for every raw H.

To continue along a physical ray, hold r, b fixed. Direct differentiation of the exact heat expression gives, with $z = H/\sqrt{n}$ and $y = (H-t)/\sqrt{n}$,

$$\partial_t \mathcal{H}_n = n\phi(z) + Ht\phi(z) - ns_n\phi(s_n y).$$

Taylor's theorem and $\|\phi''\|_\infty = \phi(0)$ bound the spatial remainder by $\phi(0)t^2/2$. Since $\partial_s[s\phi(sy)] = -\phi''(sy)$, the variance contribution is at most $\phi(0)n(s_n-1)$. Hence

$$|\partial_t \mathcal{H}_n| < \tfrac{2}{5}(t^2/2 + 1/2 + e).$$

Put $L_{r,b}(u) = \frac{2}{5}(u^2/2 + 1/2 + e) + c_0Q'_{r,b}(u)$. If Q, Q', Q'', L, L', L'' are negative at a boundary, the signs persist along the whole ray: $Q''' = -6$ and $L'' = 2/5 - 6c_0 < 0$. Thus the contact upper bound strictly decreases there. Boundary atoms and the entire closed exterior ray are excluded.

4 A continuous one-parameter certificate

For the negative ray use $z = p$ and the signed moment $-\beta$; for the positive ray use $z = 1-p$ and β. In both cases $z \in [2/5, 3/5]$, and (4) gives the directional cap

$$b_z = 2z - 1 + 2(1-z)\Theta.$$

This folds the actual probability, not the endpoints of a probability panel. It does not reflect the chosen discrepancy or change its law. Take the boundary $u = E(z + 1/2)$. Thus

$$u \in [u_0, u_1] = [2241/1250, 2739/1250], \qquad b_z = \alpha u + \delta, \quad \alpha = \frac{4963}{4980}, \quad \delta = -\frac{9889}{5000}.$$

Here $u_0^2 > 3$, so $H_4'(u) = 4u(u^2-3) > 0$; $H_4(u_1) < -2.75$ proves $H_4 < 0$ on this whole interval. Let $\mathcal{P}(u) = P_{r,\alpha u+\delta}(u)$. Replacing $|H_4|$ by $-H_4$ in (7) gives an explicit quintic $\mathcal{P}(u) = \sum_{j=0}^5 a_j u^j$, with

$$a_5 = S^3/4100, \qquad a_4 = -\tfrac{2}{5}S^3/72,$$
$$a_3 = \tfrac{2}{5}S^3/6 - c_0, \qquad a_2 = \tfrac{2}{5}S^3/12 + 51S^3/78720 + c_0(3r/2 + 3\alpha),$$
$$a_1 = -\tfrac{2}{5}S^3/2 + 3S^3/820 + 3c_0\delta, \qquad a_0 = -\tfrac{2}{5}S^3/24 - 34S^3/78720 + c_0er.$$

The exact rational endpoint inequalities are

$$\mathcal{P}'''(u_0) < -2.24, \qquad \mathcal{P}'''(u_1) < -2.27, \qquad \mathcal{P}'(u_0) > 0.915, \quad \mathcal{P}'(u_1) > 0.745.$$

Indeed $\mathcal{P}'''$ is a convex quadratic, with second derivative $120a_5 > 0$. Its negative endpoint values make it negative throughout. Therefore $\mathcal{P}'$ is concave and exceeds the chord between its positive endpoint values. It follows that $\mathcal{P}$ increases and

$$\mathcal{P}(u) \le \mathcal{P}(u_1) = -\frac{4560620039489147528734839}{25625000000000000000000000} < -0.1779. \tag{8}$$

One stronger-looking shortcut is false: $\mathcal{P}''(u_0) > 0.0228$. It is not used.

The multiplier and ray signs require a separate calculation. The boundary family

$$\mathcal{Q}(u) = Q_{r,\alpha u+\delta}(u) = -u^3 + (3r/2 + 3\alpha)u^2 + 3\delta u + er$$

has total derivatives $\mathcal{Q}''(u_0) < -1.46$ and $\mathcal{Q}'(u_1) = 179223/6250000 > 0.0286$. Since $\mathcal{Q}''$ decreases, $\mathcal{Q}'$ decreases and stays positive. Hence $\mathcal{Q}$ increases, and $\mathcal{Q}(u) \le \mathcal{Q}(u_1) < -1.198$.

For each physical ray the directional cap is fixed. Its derivatives, evaluated across the boundary family, are instead

$$\begin{aligned} q_1(u) &= -3u^2 + (3r + 3\alpha)u + 3\delta, & q_2(u) &= -6u + 3r, \\ l(u) &= \tfrac{2}{5}(u^2/2 + 1/2 + e) + c_0 q_1(u), & l_1(u) &= \tfrac{2}{5}u + c_0 q_2(u), \qquad l_2 = \tfrac{2}{5} - 6c_0. \end{aligned}$$

They are not the total derivatives of $\mathcal{Q}$. The following rational upper bounds certify their signs on the interval:

family	value at u_0	total derivative at u_0
q_1	< -4.272	< -4.452
q_2	< -7.441	$= -6$
l	< -0.903	< -1.105
l_1	< -2.330	$= -2.057$

The two quadratic families have negative constant second derivatives $q_1'' = -6$ and $l'' = -2.057$; the others are affine decreasing. Thus all four stay negative, as does $l_2 = -2.057$. Together with $\mathcal{Q} < 0$, these are the six required signs at every boundary. Holding its actual directional cap fixed, the preceding ray argument excludes the whole exterior ray beyond each boundary.

5 Completion, application and remaining premises

For each actual p the two excluded rays start at

$$u_- = E(p + 1/2), \qquad u_+ = E(3/2 - p).$$

Because $\ell > E$, the remaining support satisfies $-1/2 < Y = X/\ell + p < 3/2$ strictly. The already proved actual gap supplies

$$U_0 \le (a - A)/\ell \le d < 9/125, \qquad U_1 \ge -(b - B)/\ell \ge -d > -9/125.$$

For $\mathcal{B} = 0$ the new lower bound is $U_0 > -1/2$; for $\mathcal{B} = 1$ the new upper bound is $U_1 < 1/2$. This proves Theorem 1. If $\lambda = 0$, then $\mathbb{E}U^2 = 0$ and $U = 0$ directly. There is no division by noise or limiting argument.

Corollary 1. *Retain the accepted fixed-target entry and the ordered elementary moment/gap contraction on its 30 probability panels. On its acquired subrange $p \in [0.40, 0.53]$, all actual $n \ge 42$ with $\lambda \in [0, 3/40]$ satisfy $|U| < 1/2$ for every raw H. The previously acquired inward bound $d < 0.072 < 1/4$ is retained.*

Proof. That antecedent provides the strip $|U| < 0.6$ on the 13 closed panels 40–52 and $d \le 0.0719539227 < 0.072$. Their closed union is $[0.40, 0.53]$. Apply the theorem at the actual p. □

The new continuous calculation uses exact rational arithmetic and 28 domain/endpoint inequalities, with no new probability, phase or noise grid. Six intentionally false conditions are rejected, including the incorrect second-derivative shortcut and substitution of a total derivative for a fixed-cap ray derivative. The earlier 13-row calculation and its fresh bounded antecedent replay are preserved as additional evidence; those heat rows are not premises of this continuous proof.

The mathematical contact theorem, predecessor bound and incoming moment/gap acquisition remain separate premises. This result neither certifies a finite base nor supplies strict count exclusion. It does not remove any existing certificate or assert full-noise confinement on all 13 panels. The already separate full-noise result on $[0.41, 0.42]$ is unchanged.